

Article

Deep Reinforcement Learning-Guided Bio-Inspired Active Flow Control of a Flapping-Wing Drone for Real-Time Disturbance Suppression

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Abstract

Flapping-wing drones (FWDs), owing to their compact size and operation in cluttered and unsteady airflow environments, encounter significant aerodynamic and stability challenges. Studies of avian flight reveal that falcons and other raptors actively deflect their covert feathers to mitigate gusts and maintain stable flight. Drawing inspiration from this mechanism, this study presents a peregrine falcon-inspired Active Flow Control Unit (AFCU) integrated with a Deep Deterministic Policy Gradient (DDPG)-based deep reinforcement learning (DRL) controller for real-time disturbance attenuation. The AFCU employs mechanical covert feathers (MCFs) that actuate to dissipate gust loads during high wind conditions. A reduced-order bond graph model that encapsulates the nonlinear interaction between the primary wing and the feather-based active flow control surfaces is created which is used as a dynamic training environment for the DDPG agent. Utilizing closed-loop interactions, the successfully obtained learned policy produces optimal actuator forces to reduce feather-displacement error and aerodynamic load variations. The designed controller stabilizes the internally unstable open-loop AFCU, attaining near-zero steady-state error and settling times under 1.6 s for gust magnitudes ranging from 12.5 to 20 m/s. Simulations further illustrate a reduction of up to 50% in gust-induced loads compared to traditional approaches. This integration of bio-inspired design with learning-based active flow control offers a viable avenue for the development of highly adaptive and gust-resilient flapping-wing aerial systems.

**Keywords:** DDPG; DRL control; active flow control; bio-inspired; flapping wing; drone

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1. Introduction

Maintaining a stable attitude is particularly challenging for unmanned aerial vehicles (UAVs) operating in the atmospheric boundary layer, where turbulence levels are inherently high. Within this region, unmanned aircraft are frequently subjected to sudden and intense gusts that can rapidly disrupt their orientation, position, and airspeed, demanding highly resilient active gust suppression systems and advanced control strategies [1–3].

A wide range of conventional disturbance mitigation strategies for drones has been extensively investigated. Ren et al. [4] introduced a pressure-sensor-based gust-mitigation

method that uses real-time flow measurements to provide phase-advance information unavailable to standard inertial sensors. Disturbances estimated from two wing-mounted sensors were fed into a proportional integral derivative (PID) controller to counteract gust-induced rolling moments. Their simulations showed that incorporating flow-field data improves gust rejection compared to conventional approaches. However, the method was demonstrated only on fixed-wing drones, and its applicability to flapping-wing platforms remains unexplored.

Recent work shows that synthetic jet actuators can lower aerodynamic drag in UAVs and suppress gust-induced oscillations, thereby enhancing overall stability [5]. Similarly, vortex generators have been investigated for their ability to reduce wing loads and delay wing-tip stall during turbulent, high-speed flight [6]. Despite these advances, such gust-alleviation technologies have been tested mainly on fixed-wing platforms, and their effectiveness for FWDs remains largely unexplored. Another study [7] introduced a turbulence-mitigation method that reduces gust detection delay and enhances the responsiveness of a UAV's onboard control electronics. By applying advanced signal-processing algorithms to MEMS gyroscope data, the authors achieved improved stability during small-scale gust events.

Several bio-inspired flying robot designs have also been studied in depth to address turbulence [8–12]. A recent work [13] proposes a bio-inspired “fly-by-feel” scheme that enables flapping-wing UAVs to sense aerodynamic loads through wing-embedded strain sensors, mimicking insect mechanoreceptors. Using reinforcement learning (RL) to interpret these strain signals, the system performs real-time estimation of both attitude and airflow without depending on inertial sensors. Tests conducted in simulations and flight experiments with wind disturbances of 3–7 m/s demonstrated reliable and adaptive flight behavior. Nonetheless, the approach has not yet been assessed under more intense turbulence conditions beyond 7 m/s.

Drawing inspiration from avian feathers, some previous studies have explored bio-inspired feathered wing configurations for improving gust mitigation in flapping-wing UAVs. For example, the study in [14] introduced a bio-inspired feathered wing design for flapping-wing UAVs aimed at dissipating gust effects in turbulent airflow. Their simulations showed that wings equipped with feathers managed gust disturbances more effectively than plain wings.

Beyond active gust-alleviation design approaches, a substantial body of work has explored the mitigation of gust effects in flapping-wing drones through both linear and nonlinear control strategies [15–19]. A neural network-based hybrid mode-switching controller has been proposed [20] for three-dimensional position stabilization of FMAVs. The approach combines a novel vertical-frame dynamic formulation with a radial basis function (RBF) neural network-based adaptive controller, allowing the system to switch between control modes to avoid unsafe flight conditions and improve efficiency. Learning updates are adjusted during mode transitions, and saturation together with barrier Lyapunov functions are used to constrain lateral velocity. The closed-loop system is shown, via Lyapunov analysis, to be semi-globally uniformly ultimately bounded. Experimental results validate the reliability and effectiveness of the proposed strategy.

A wing kinematics-based deep reinforcement learning (DRL) controller was developed to stabilize FMAVs under large 6-DOF disturbances. The control policy was trained using the Soft Actor–Critic (SAC) algorithm in a high-fidelity computational fluid dynamics (CFD)-coupled simulation environment. The results showed robust and rapid stabilization during hovering under severe perturbations [21].

Another research presents a multi-layer model predictive control (MPC) scheme that enhances control accuracy and computational efficiency in flapping-wing MAVs. The

method uses an MPC with an extra layer that adjusts wing kinematics. The results indicate that this approach delivers faster and more precise tracking than typical MPC controllers when exposed to gust disturbances [22]. However, its performance declines under stronger gusts, highlighting the need for further robustness improvements.

An LQR controller is utilized by the authors in [23] to accomplish trajectory tracking in an insect robot traveling at up to 25 cm/s, but the work only examined minor disturbances and has not evaluated gust-rejection performance. Later, LQR was employed by Bhatia et al. [24] to control FWD attitude in turbulent airflow. The study showed that although the platform was inherently unstable in open loop, closed-loop LQR feedback was essential for accurate attitude stability and sustaining the flight while operating in gusts. The controller successfully controlled gust disturbances up to 3 m/s.

A tailless flapping drone with three wing pairs and bionic elastic legs was studied by Zheng et al. [25], allowing for a smooth transition between flight and ground movement. In the proposed system, flight stabilization and maneuvering are achieved primarily through asymmetric wing kinematics and differential flapping control rather than through a conventional tail surface. Stable hovering and accurate trajectory control were achieved by combining a cascaded PID controller with an SE (3) geometric tracking method. The robot attained adequate stability and maneuverability despite the yaw axis remaining underactuated.

While many of the current control solutions work well in situations with mild disruptions, they become less successful in situations with greater gusts. This restriction results from the fact that gust rejection is frequently handled as a stand-alone control issue without incorporating adaptive aerodynamic techniques. For flapping-wing robots operating in highly nonlinear and rapidly fluctuating airflow, single-controller designs are insufficient according to a recent survey study [26], which highlights the necessity of active aerodynamic elements or multi-layer control frameworks. This gap led the researchers to explore natural flyers, where they found that birds use intermittent flight methods in turbulent situations, where the passive deployment of covert feathers helps attenuate rapid gust loads [27]. Figure 1 shows the flight of a falcon during turbulence where the covert feathers are deflected.

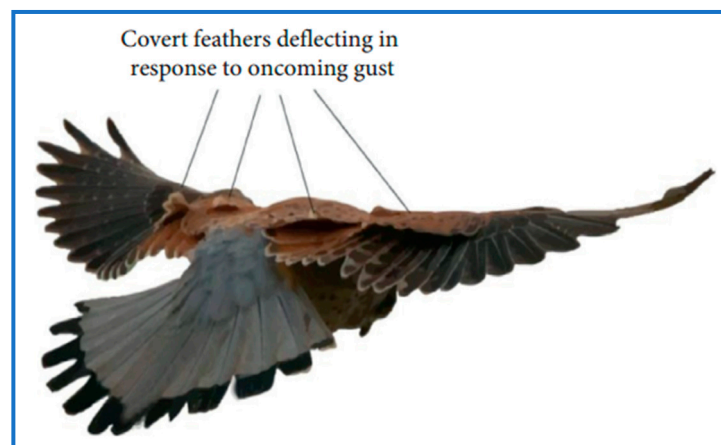


Figure 1. Coverts deflection (reproduced from [28]).

Authors in [29,30] have further shown that bio-inspired feathered flapping-wing UAVs and ornithopters are dynamically unstable under gusts, requiring active control systems to achieve flight stability and robustness under gusty conditions. Further enhancing these results, this study proposes a peregrine falcon-inspired Active Flow Control Unit (AFCU) equipped with mechanical covert feathers (MCFs) and controlled through a Deep Deter-

ministic Policy Gradient (DDPG)-based deep reinforcement learning (DRL) framework. The main contributions of this study are as follows:

- Development of a peregrine falcon-inspired AFCU comprising MCFs and modeled through a reduced-order bond graph model that captures the nonlinear coupling between the main wing and feathered flow-control surfaces.
- Design and training of a DDPG-based DRL controller, eliminating the need for predefined control laws.
- Successful stabilization of the internally unstable open-loop AFCU, achieving near-zero steady-state error and settling times below 1.6 s across gust magnitudes of 12.5–20 m/s.
- Validation through simulation and literature comparison to confirm the adaptability and aerodynamic resilience of the proposed control framework.

The rest of the paper is structured as follows: The design and modeling of the FWD wing with the AFCU are covered in Section 2. The stability study is presented in Section 3, and the design of the DDPG-based DRL controller is described in Section 4. The simulation findings are covered in Section 5, and the study is concluded in Section 6.

2. Design and Bond Graph Modeling of AFCU

This work takes the Smart Bird developed by Festo [31] as the prototype flapping ornithopter. The developed AFCU configuration includes 16 MCFs, arranged with eight on the upper side of the wing and eight on the underside. A flap, piezoelectric transducer (PZT), voice-coil actuator, mechanical linkage, spring and bio-inspired controller are all included in each MCF. Figure 2 shows the MCF's configuration and Figure 3 displays the block diagram of the MCF's operation and energy flow.

A compact voice-coil actuator is used to actuate each mechanical covert feather because of its quick response time and accurate displacement control. Voice-coil actuators are frequently employed in applications that call for high bandwidth, smooth linear motion, and a very straightforward mechanical structure. The linked electromechanical behavior of the feather actuation system is represented in the proposed AFCU architecture by incorporating the actuator dynamics into the bond graph formulation. The control properties of coordinated feather actuation and the system's subsequent gust-mitigation performance are the main topics of this work. Future prototype development of the AFCU idea will address specific actuator dimensions, including mass, power requirements, and hardware implementation constraints.

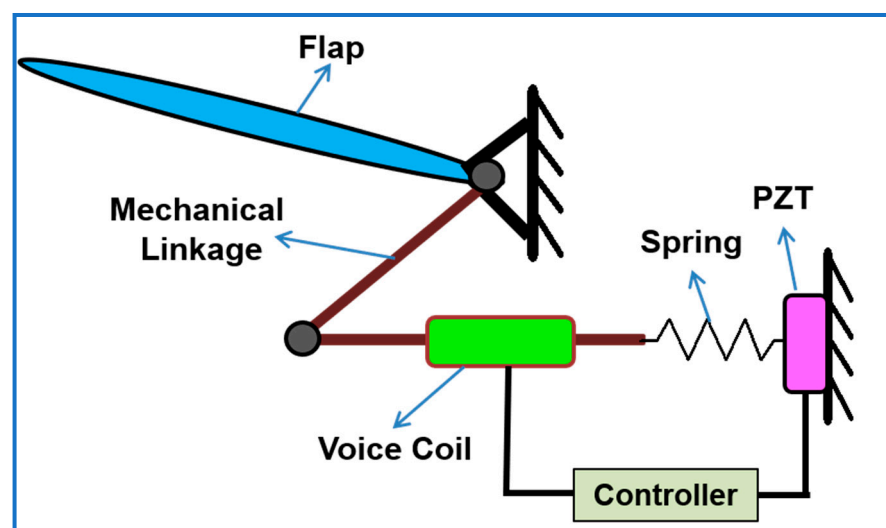


Figure 2. Mechanical covert feather (MCF).

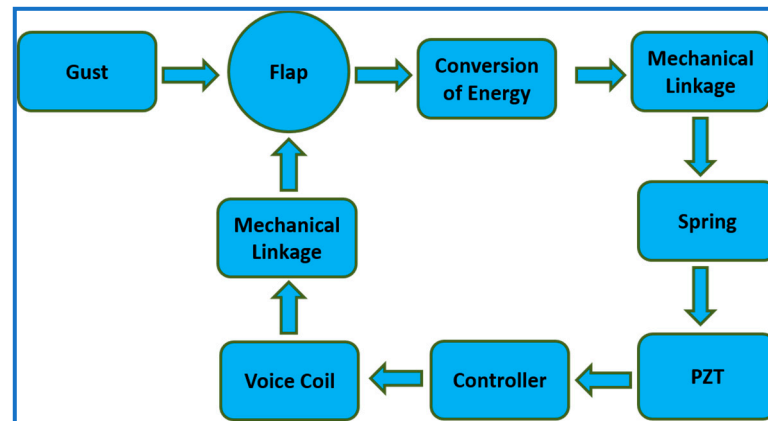


Figure 3. Block diagram of the MCF.

Bond graphs are a flexible, energy-based modeling framework that may cohesively describe the power interactions and dynamic behavior of physical systems. Because of this, they are especially helpful for flapping-wing UAVs, which need to accurately capture the interplay between actuators, mechanical linkages, and unstable aerodynamic stresses. Bond graphs can also be readily converted into state-space models, permitting direct engagement with optimization, simulation, and control design tools. Additionally, because of their modular design, researchers can construct incremental models by improving individual subsystems without changing the overall system architecture. When adding bio-inspired elements, like MCFs or compliant mechanisms, to an already-existing FWD design, this capacity is especially helpful [32].

Bond graphs provide a reliable modeling framework for AFCU-type bio-inspired systems since they have been extensively used and experimentally confirmed in previous ornithopter and flapping-wing investigations [33]. As shown in Figure 4, the modeling process starts with building the bond graph of a single MCF, then the wing model, and then integrating all sub-components into a complete bond graph representation of the wing installed with all 16 MCFs. The parameters used in the bond graph are taken from the authors' previous studies [29,30]. Finally, the equations of a single MCF obtained via the bond graph model are as follows:

$$\dot{p}_1 = ic \cdot p_3 + ic \cdot q_3 \quad (1)$$

$$\dot{q}_1 = \frac{1}{I_1} \cdot p_2 \quad (2)$$

$$\dot{p}_2 = \frac{ic}{l} p_3 + \frac{ic}{l} q_3 - \frac{1}{C} q_1 - \frac{1}{C_1} q_2 - \frac{m}{C_2} q_4 \quad (3)$$

$$\dot{q}_2 = \frac{1}{I_1} \cdot p_2 \quad (4)$$

$$\dot{p}_3 = q_5 \quad (5)$$

$$\dot{q}_3 = S_f - \frac{1}{l \cdot I_1} p_2 - \frac{1}{l} p_1 \quad (6)$$

$$\dot{q}_4 = \frac{m}{I_1} p_2 - \frac{R}{C_2} q_4 \quad (7)$$

$$\dot{q}_5 = \frac{1}{l \cdot I_1} p_2 \quad (8)$$

where q_1, q_2, q_3 , and q_4 represent the generalized displacements of the compliance elements and p_1, p_2 , and p_3 depict the generalized momentum of inertia elements. The overall system

model contains 134 differential equations because of the 134 energy-storing elements of the bond graph in Figure 4.

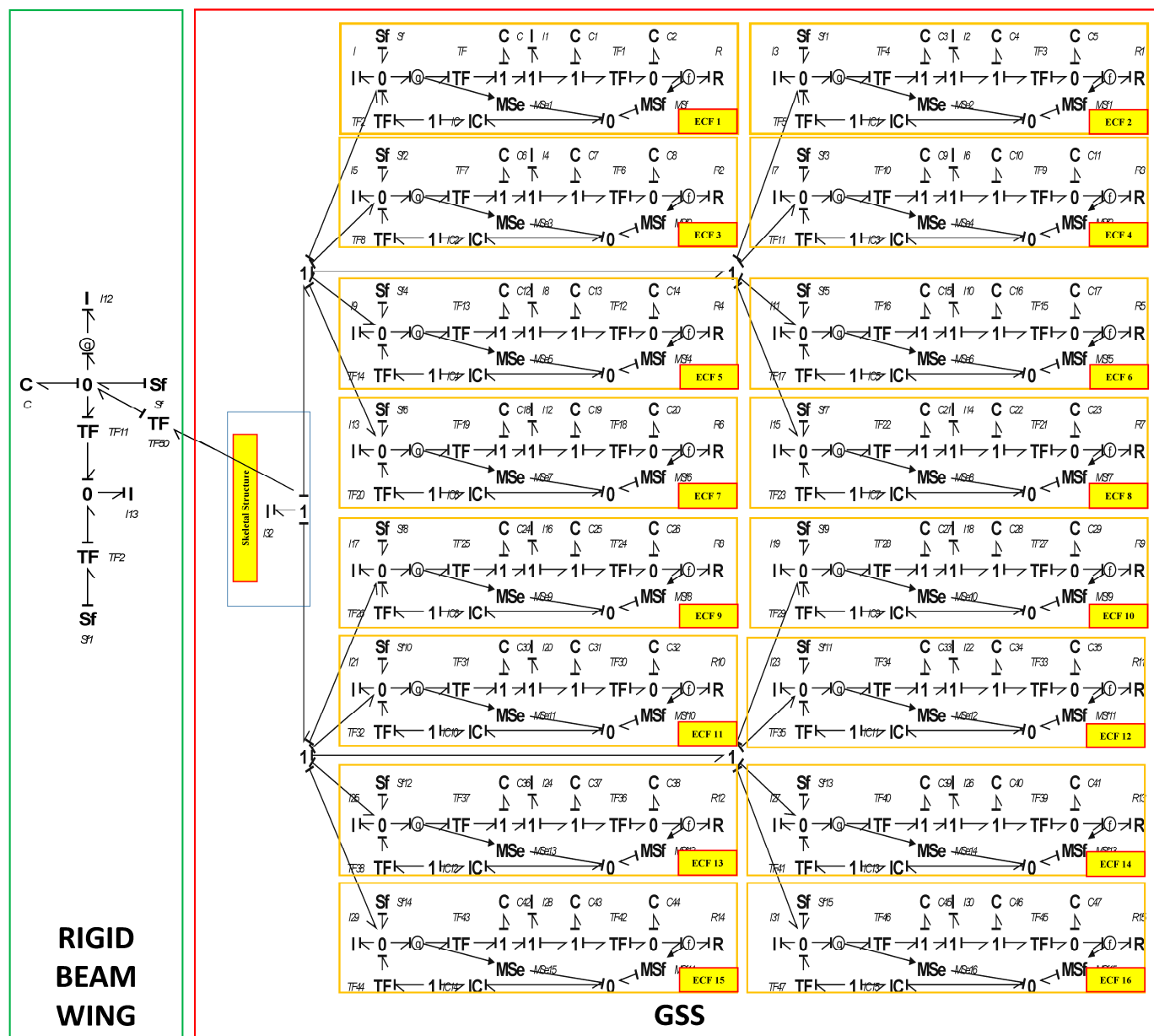


Figure 4. AFCU-installed wing model based on bond graph.

Standard unsteady aerodynamic approximations are used to express the aerodynamic effort sources in the bond graph model in order to improve the aerodynamic depiction of the airflow–feather interaction [22,32]. The quasi-steady lift force produced under a gust for each MCF is determined by

$$F_L = \frac{1}{2} \rho A C_L(\alpha) (V_g + \dot{q}_f)^2 \quad (9)$$

where ρ is air density, A is the feather area, $\dot{q}_f$ represents the flap tip velocity modified by the incoming gust velocity V_g and α is the angle of attack [26,32]. The lift coefficient is now linearized as

$$C_L(\alpha) = C_{L\alpha} \quad (10)$$

The equivalent moment at the MCF is

$$M_a = F_L l_f \quad (11)$$

where the moment arm is defined as l_f . The drag because of the gust is likewise modeled as

$$F_D = \frac{1}{2} \rho A C_D (V_g + \dot{q}_f)^2 \quad (12)$$

Additionally, gust-structure coupling is used to represent an equal aerodynamic disturbance force:

$$F_g(t) = \rho A V_g(t) \dot{q}_f \quad (13)$$

We now use Balanced Truncation to get a reduced-order model (ROM) of the AFCU-incorporated FWD wing using Hankel Singular Values [34]. We perform this reduction solely to obtain a physics-faithful yet computationally tractable representation of the AFCU dynamics. The retained states correspond to the dominant energy-storing modes, ensuring that the reduced system reproduces the frequency response and gust-interaction behavior of the full 134-state model with minimal error. Overall, the ROM successfully retains the dominant system dynamics. The final reduced seventh-order state vector is $\mathbf{x} = [q_{f1} \ q_{f3} \ q_{f4} \ q_{f6} \ q_{f7} \ q_{f8} \ q_{f9}]^T$ which contains generalized displacements of respective feather flaps. The final state-space can hence be stated in a general form as:

$$\begin{aligned} \dot{\vec{x}}(t) &= A \cdot \vec{x}(t) + B \cdot \vec{u}(t) + B_d \cdot \vec{w}(t) \\ \vec{y}(t) &= C \cdot \vec{x}(t) + D \cdot \vec{u}(t) \end{aligned} \quad (14)$$

where B_d indicates gust influence and $\vec{w}(t)$ represents wind gust. Hence, the reduced-order state space model of the AFCU-equipped FWD wing is:

$$A = \begin{bmatrix} -8.8 & -2.5 & -3.3 & 1.8 & 5.1 & -3.1 & -6.1 \\ -9.9 & -1.2 & -9.1 & 2.05 & 2.6 & -4.2 & -6.7 \\ -5.03 & 7.5 & -4.7 & 3.5 & -4.1 & 0 & -7.6 \\ -9.3 & 3.3 & -2.9 & -1.5 & -6.5 & 6.1 & 4.9 \\ 3.1 & 4.1 & -3 & -1.9 & -8.1 & 3.4 & 6.4 \\ 0 & -6.4 & 0 & 7.2 & 3.3 & -0.04 & -2.9 \\ -4.4 & 0.4 & -1.4 & 1.2 & -0.7 & -3.5 & -4.03 \end{bmatrix}$$

$$B = [2.9 \ 3.2 \ 0 \ -3.4 \ 0.21 \ 0 \ -0.9]^T$$

$$C = [0.07 \ 4.32 \ 0 \ -0.06 \ 0.1 \ 0 \ 1.9] \quad D = [0]$$

As with most control-oriented aerodynamic models, several simplifying assumptions are adopted to maintain tractability. High-fidelity effects such as broadband stochastic gusts, actuator saturation, sensor noise, and fine-scale feather couplings are not explicitly represented in the present formulation. These factors do not materially affect the dominant FSI modes captured in the bond graph model or the closed-loop behavior evaluated in this study, but they may influence hardware-level implementation. Future work will integrate actuator bandwidth limits and stochastic gust models for higher-fidelity validation.

We further mention here that the current model uses a quasi-steady formulation to represent the aerodynamic forces. This method is frequently used in control-oriented modeling of flapping-wing systems because it captures the dominant aerodynamic trends while maintaining tractable system dynamics, despite ignoring some unsteady aerody-

dynamic mechanisms related to flapping flight, such as leading-edge vortex evolution, wake interaction, and added-mass effects. Rather than reproducing the entire complexity of flapping aerodynamics, the main goal of this study is to assess the viability of reinforcement learning-based control for the proposed active flow-control architecture and study its benefit for gust suppression. As a result, the reduced-order model maintains the primary aerodynamic–structural interaction important to gust mitigation while providing a computationally effective training environment.

3. Open-Loop Stability

The open loop poles of the AFCU-wing system are located at -5.8 , -20 , -0.91 , 6.4 , -2.6 and $-2.6 \pm 5.3j$. We notice that the pole at 6.4 is in the right-half-plane which shows that the AFCU-equipped FWD wing system is intrinsically unstable. Additionally, it is also noted that when gust disturbances are applied, the reduced-order AFCU model's open-loop state response diverges. This behavior results from the linearized description of the aeroelastic system, where there is no active feedback mechanism to control the motion of the feather and aerodynamic loading functions as an external excitation. Unbounded reactions in simulation can result from the aerodynamic forcing exciting unstable structural modes of the coupled aero-mechanical system in the absence of control. Therefore, this behaviour is a reflection of the intrinsic instability of the untrained system and motivates the design of a learning-based controller.

4. DDPG-Based DRL Controller

We now synthesize the DDPG-based DRL controller for the AFCU installed wing system. The objective of the DDPG agent is to learn a control policy that can suppress gust-induced perturbations by modulating the input effort acting on the feather mechanical linkage. The learned policy replaces traditional gain-scheduled or model-based controllers and adapts autonomously to changing aerodynamic conditions without requiring online model updates or explicit disturbance estimation.

4.1. Deep Deterministic Policy Gradient (DDPG)

An actor–critic method appropriate for continuous control serves as the foundation for the DRL framework in this study. Figure 5 illustrates the DDPG framework utilized in this work. The tuple defines the learning process as a Markov Decision Process (MDP) [35,36] as follows:

$$M = (S, A, T, r, \gamma) \quad (15)$$

where S and A represent the state and action spaces, $T(s_{t+1} | s_t, a_t)$ is the transition probability, $r(s_t, a_t)$ is the reward, and $\gamma \in (0,1)$ is the discount factor. The objective of the deterministic policy $\mu(s_t | \theta^\mu)$ is to maximize the expected cumulative return [35,36]:

$$J(\theta^\mu) = E_{s_t \sim \rho^\mu} \left[\sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, \mu(s_{t+k})) \right] \quad (16)$$

The agent is trained by minimizing the temporal-difference (TD) error between the predicted and target Q-values [35,36]:

$$L(\theta^Q) = E_{(s_t, a_t, r_t, s_{t+1})} \left[\left(Q(s_t, a_t | \theta^Q) - y_t \right)^2 \right] \quad (17)$$

where the TD target y_t is computed as [35,36]

$$y_t = r_t + \gamma Q'(s_{t+1}, \mu'(s_{t+1} | \theta^{\mu'}) | \theta^{Q'}) \quad (18)$$

Here, (μ', Q') represent the slowly updated target networks. The target parameters are updated by soft replacement [35,36]:

$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta' \text{ where } \tau \ll 1 \quad (19)$$

The actor network updates according to the deterministic policy gradient method [35,36]:

$$\nabla_{\theta^{\mu}} J \approx E_{s_t} \left[\nabla_a Q(s_t, a | \theta^Q) \big|_{a=\mu(s_t)} \nabla_{\theta^{\mu}} \mu(s_t | \theta^{\mu}) \right] \quad (20)$$

Through alternating updates of the actor and critic, the policy converges toward the optimal control mapping that maximizes the long-term reward. The DDPG algorithm combines the value-based stability of Q-learning with the gradient-based adaptability of policy optimization [35,36].

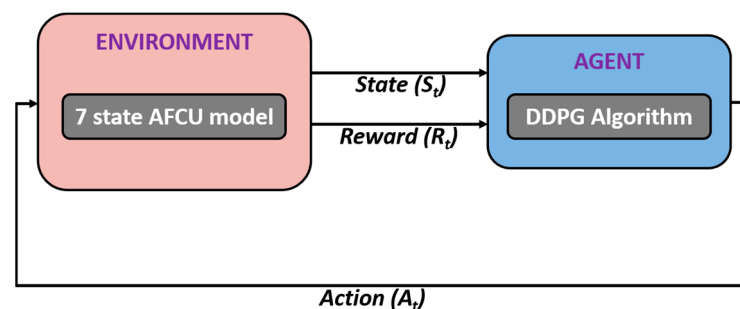


Figure 5. DDPG framework used for AFCU control.

The DDPG framework employs an actor–critic structure that performs policy evaluation and improvement in parallel. The actor network $\mu(s_t | \theta^{\mu})$ generates a continuous control signal corresponding to the current state of the feathers, while the critic network $Q(s_t, a_t | \theta^Q)$ estimates the expected return associated with each action. Both networks share a common state input but have separate parameter sets. Each consists of two fully connected ReLU-activated hidden layers (64×64 neurons), and soft target networks (μ', Q') are employed to enhance training stability. This dual-network configuration enables smooth policy updates and continuous control generation suited for gust mitigation in the FWDs.

4.2. Control Objective

The primary control objective is to preserve bounded control effort and structural safety while achieving stable regulation of the AFCU under varying gust intensities. Under gust magnitudes up to 20 m/s, the DDPG agent is trained to reduce overall state deviation, inhibit overshoot, and push all system states toward their steady-state equilibrium. Every training episode starts with a perturbed initial condition and exposes the system to step-type gust disturbances that activate the AFCU’s seven dynamic modes. The controller’s job is to quickly counteract these gust effects and bring the system back to stability without going over actuator saturation. Effective disturbance rejection and energy-efficient control are measured by smooth convergence of all states to the steady state in around 1.7 s with control forces staying below 0.8 N.

4.3. Training Setup for DDPG Controller

The DDPG environment is custom-built in Python following the OpenAI Gym (version 0.26.2) design to enable direct interaction between the learning agent and the reduced-order AFCU dynamics. The environment supports on-policy and off-policy algorithms and was interfaced with a PyTorch (version 2.5.1) -based DDPG framework for continuous

action learning. This modular setup ensures reproducibility of training, testing, and policy evaluation under varying gust conditions.

The reduced-order model of the AFCU provides a computationally efficient dynamic approximation for the reinforcement learning environment. While the full 134-state bond graph model accurately represents the physical system, training a DDPG agent on such a high-dimensional state space is computationally prohibitive and leads to poor sample efficiency and unstable gradient updates. The DDPG policy does not rely on the ROM for analytical controller synthesis; rather, it interacts with it to learn closed-loop behavior. The seventh-order ROM retains all dominant MCF–wing coupling dynamics while enabling fast, stable simulation over long training horizons.

4.4. Reward Formulation

The reinforcement signal is designed to balance state convergence, control efficiency, and output oscillations within the AFCU dynamics. The agent obtains a scalar reward computed from a weighted combination of state deviation, control effort, and output constraints after each time step. Specifically, quadratic penalties are applied to the overall state energy ($\|x\|^2$) and control input (u^2), along with a proportional term on the output magnitude ($|y|$) to discourage excessive oscillations. The complete reward formulation is expressed as

$$r = -\left(0.1 \|x\|^2 + 0.01 u^2 + 0.1 |y|\right) \quad (21)$$

The quadratic state energy term $\|x\|^2$ implicitly directs all seven state variables toward their equilibrium values, thereby enforcing overall system stabilization. In addition, the control input is constrained within the range $[-0.8, 0.8]$ N through the environment's continuous action space, thereby enforcing a hard actuator saturation limit during both training and evaluation.

4.5. Training Hyperparameters

The actor–critic architecture is implemented in PyTorch using two fully connected hidden layers of 64 neurons each for both networks. The agent is trained using the Adam optimizer to ensure stable convergence. Key training parameters are empirically tuned to achieve a balance between learning stability and responsiveness. The complete set of hyperparameters used in training the DDPG-based DRL controller is given in Table 1.

Table 1. Hyperparameters for the DDPG-based DRL controller.

Description	Symbol	Value
Discount factor	γ	0.99
Critic learning rate	α_c	1×10^{-4}
Actor learning rate	α_a	1×10^{-5}
Replay buffer size	N	10,000
Soft update rate (target networks)	τ	0.005
Hidden layer	l	64×64
Mini-batch size	k	64
No of control outputs	n_u	1
No of state variables	n_x	7
Maximum steps per episode	t	Environment defined (≈ 300 –600)
Total training episodes	K	500

4.6. Computational Setup

The DDPG agent is trained for 500 episodes, each consisting of 300–600 simulation steps. The training is executed in Python 3.10 using the PyTorch framework on an Intel® Core™ i7 processor (2.9 GHz) and 16 GB RAM. Average training time per episode is approximately 3–4 s, and convergence of the reward curve is typically achieved after 400–450 episodes. The trained actor network is subsequently used for closed-loop evaluation and visualization of state trajectories at various gust intensities.

In addition, to assess performance under time-varying disturbances, a curriculum-based training approach is used. The agent is first trained with step-gust inputs and then exposed to sinusoidal gust profiles. This gradual shift from simple to more complex disturbances helps the controller handle different gust patterns more effectively, without changing the reward function or training parameters.

Although the DDPG-based controller provides stable gust-suppression behavior in simulation, several limitations remain. The training relies on a reduced seventh-order linear model, which omits nonlinear feather–airflow interactions and may not fully represent real flapping-wing aerodynamics. DDPG itself is sensitive to hyperparameters and may converge to suboptimal policies, especially under noisy or sparse rewards. Furthermore, actuator saturation dynamics, sensor noise, and communication delays are also not included in the training environment. These factors may influence real-world performance and motivate further development using higher-fidelity models and hardware-in-the-loop testing.

5. Simulation Outcomes and Insights

After completing the training phase in the custom OpenAI Gym-based environment, the learned actor policy is extracted and deployed for performance validation in this section. The evaluation involves applying the trained agent, originally optimized for a 15 m/s gust, to unseen gust magnitudes of 12.5, 17.5, and 20 m/s. Each test run simulates a step-gust disturbance and observes the state's response. The controller's objective is to ensure smooth convergence of all the states to their steady-state equilibrium under gust disturbances within less than 1.7 s while minimizing control effort and maintaining overall system stability.

The resulting state trajectories obtained from these evaluations are presented in Figures 6–9. The overall dynamic behavior of the AFCU system under these four gust levels is analyzed below to illustrate the controller's adaptability and consistency across varying aerodynamic conditions. At the lowest gust intensity of 12.5 m/s, the DDPG-based controller demonstrates highly damped behavior with rapid decay of transient components as shown in Figure 6. The reduced aerodynamic coupling among the MCFs allows the system to settle smoothly with minimal oscillation.

When exposed to the 15 m/s gust the controller shows the most optimal and consistent output (Figure 7). The transient peaks of all the state variables are nearly identical, and the settling times remain within approximately less than 1.4 s. Figure 8 shows that the DDPG agent continues to create stable trajectories with somewhat longer settling times under a larger gust of 17.5 m/s. In the mid-range, little oscillations appear, but these are soon suppressed as the learnt feedback action makes up for the increased aerodynamic loading. The policy effectively generalizes outside of its training environment, as seen by the overall profile that remains well-damped. As seen in Figure 9, the system still displays a stable response and full recovery to the steady state at the extreme gust level of 20 m/s. Transient deviations are more noticeable at first, but the controller quickly returns to equilibrium without divergence or overshoot amplification.

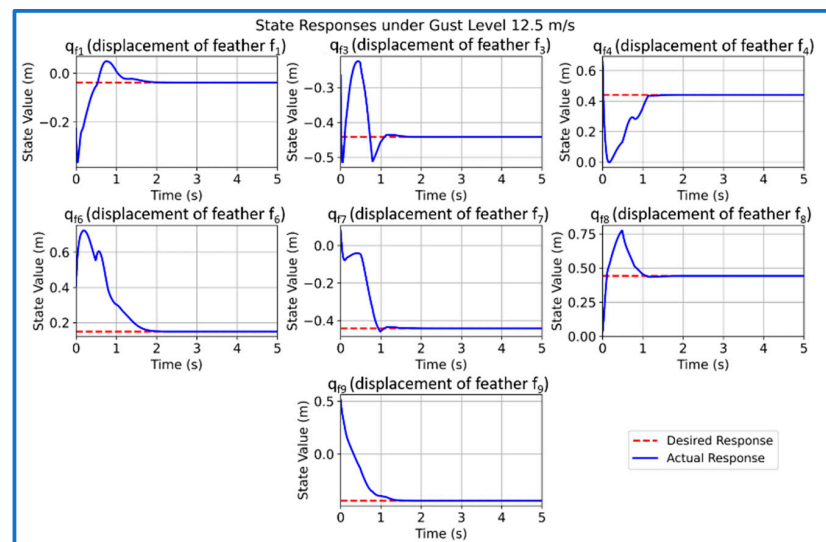


Figure 6. DDPG-controlled state response at 12.5 m/s gust.

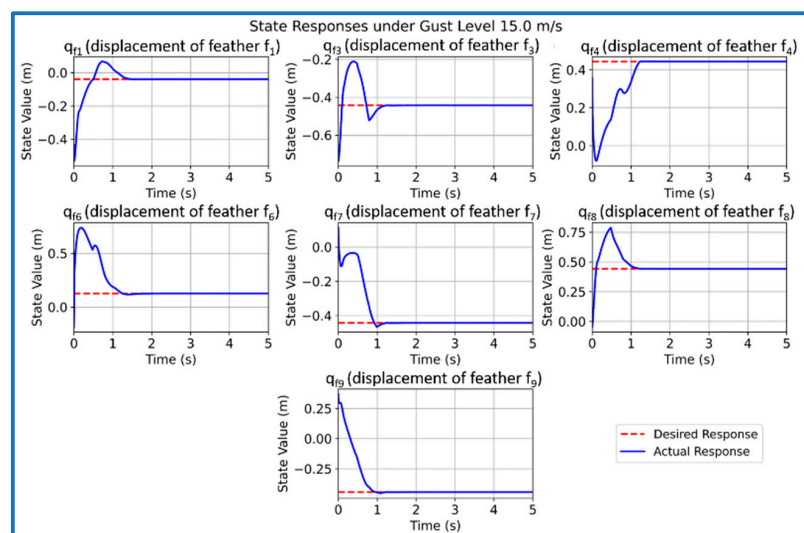


Figure 7. DDPG-controlled state response at 15 m/s gust.

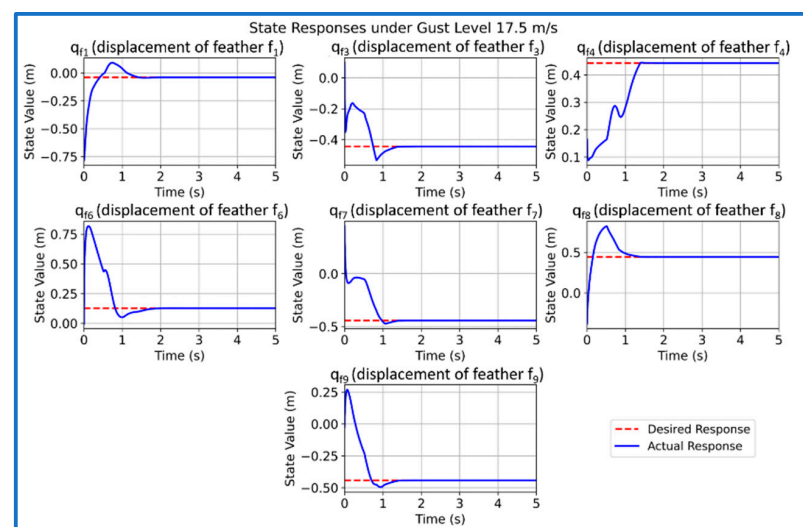


Figure 8. DDPG-controlled state response at 17.5 m/s gust.

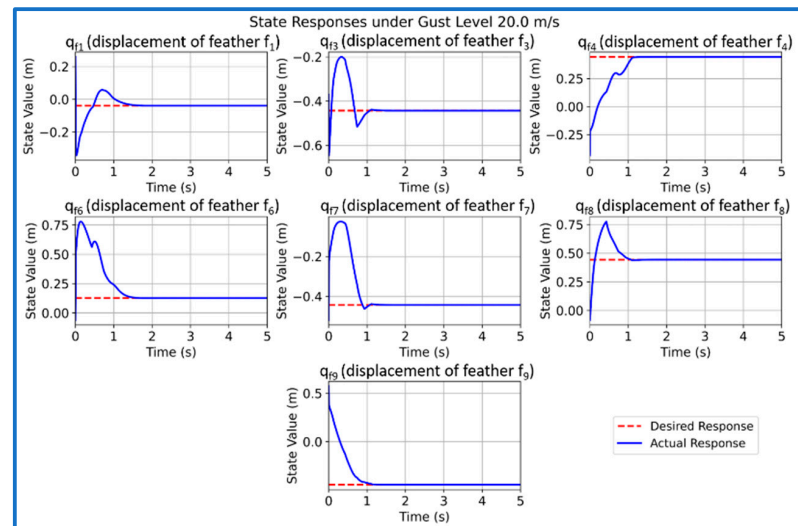


Figure 9. DDPG-controlled state response at 20 m/s gust.

According to a comparative analysis of all the gust situations, the DDPG agent maintains a similar stabilization pattern across all the disturbance levels. The qualitative character of the state responses does not change with gust magnitude, although the transient amplitudes and settling times do. Instead of confining the learning to case-specific dynamics, this consistency proves that the trained policy has successfully discovered a mapping between aerodynamic perturbations and corrective control actions. In this way, the controller achieves both robustness and generalization, which are crucial characteristics for dependable gust-mitigation performance in actual flying situations.

The respective control input responses of the DDPG agent at four gust scenarios are presented in Figure 10, which show that the control input behavior remains smooth, stable, and energy efficient. Even at the extreme 20 m/s gust, the control force converges rapidly after a brief transient and stays within the ± 0.8 N actuator limit. Overall, the comparison across Figure 10 confirms that the reward formulation effectively regulates efficient energy usage. Thus, the quadratic control-cost term and hard action-space constraint together ensure smooth, bounded control effort, validating that the learned policy preserves both efficiency and actuator safety under varying gust intensities.

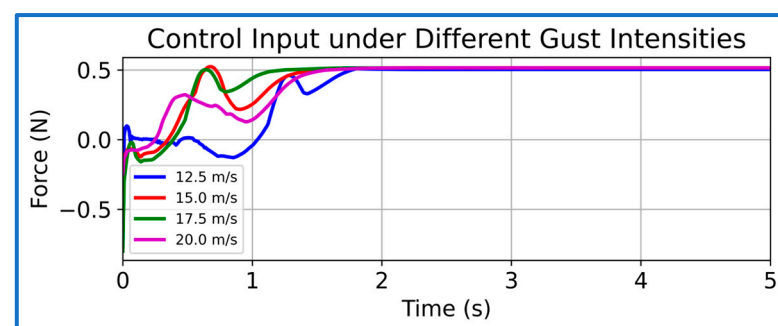


Figure 10. Control input plots of DDPG controller at different gust speeds.

The results summarized in Table 2 indicate that the DDPG controller successfully settles all the states within approximately less than 1.6 s and negligible steady-state error. Minor variations in rise times correspond to the dynamic coupling among adjacent feathers.

Table 2. State response comparison of the DDPG controller: Rise time (RT) and settling time (ST).

States (Displacement of Feathers)	12.5 m/s		15 m/s		17.5 m/s		20 m/s	
	RT (s)	ST (s)	RT (s)	ST (s)	RT (s)	ST (s)	RT (s)	ST (s)
q_{f1}	0.02	1.56	0.01	1.29	0.00	1.28	0.01	1.42
q_{f3}	0.03	1.02	0.00	1.05	0.79	1.27	0.01	0.98
q_{f4}	0.00	1.12	1.11	1.19	1.27	1.36	0.00	1.10
q_{f6}	0.08	1.58	0.10	1.18	0.05	1.57	0.07	1.36
q_{f7}	0.89	1.03	0.89	1.09	0.00	1.29	0.00	1.00
q_{f8}	0.35	0.99	0.34	1.05	0.34	1.21	0.31	0.96
q_{f9}	0.00	1.31	0.81	1.09	0.71	1.28	0.00	1.07

Error characterization of the trained DDPG-based AFCU controller under a 15 m/s gust is discussed here. The histogram of normalized feather-displacement errors in Figure 11 shows that more than 85% of all the samples lie within 20% error, indicating precise steady-state tracking of the desired feather deflections. The right-skewed distribution reflects transient deviations that occur only during gust onset and rapidly diminish thereafter. The temporal evolution of the mean state error in Figure 12 demonstrates a smooth exponential-like decay, where the displacement error decreases from 100% to below 5% within approximately 1.6 s. These results confirm that the proposed DDPG-based AFCU controller achieves stable convergence and localized feather coordination, maintaining feather-level errors within acceptable aerodynamic tolerance ($\leq 20\%$) during sustained gust conditions.

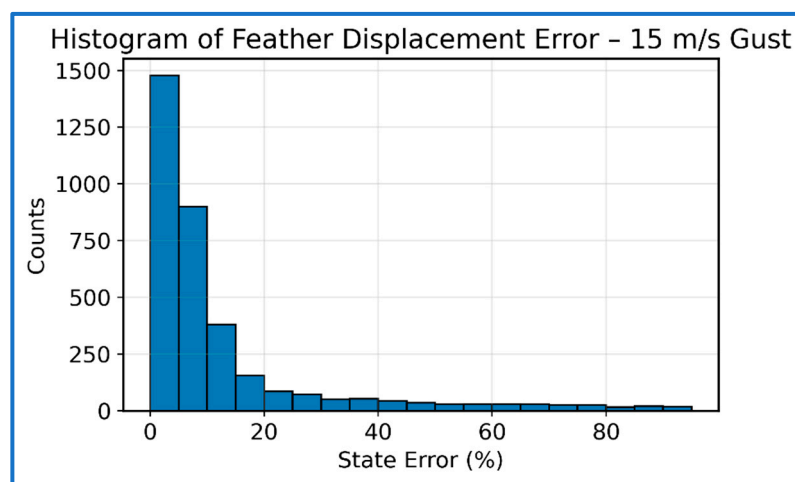
**Figure 11.** State errors at 15 m/s gust.

Figure 13 presents the step responses and control efforts of the DDPG agent at various gusts. We noticed that the controller exhibits a rapid rise and near-identical settling behavior with settling times less than 1.6 s. The negligible rise times at higher gusts reflect the agent's anticipatory control learned through training, allowing near-instantaneous compensation. In contrast, the untrained AFCU wing system was inherently unstable as the open-loop step response was diverging. The trained policy successfully stabilizes this previously unstable system, maintaining bounded AFCU dynamics. The details of the DDPG-controlled step plots are summarized in Table 3.

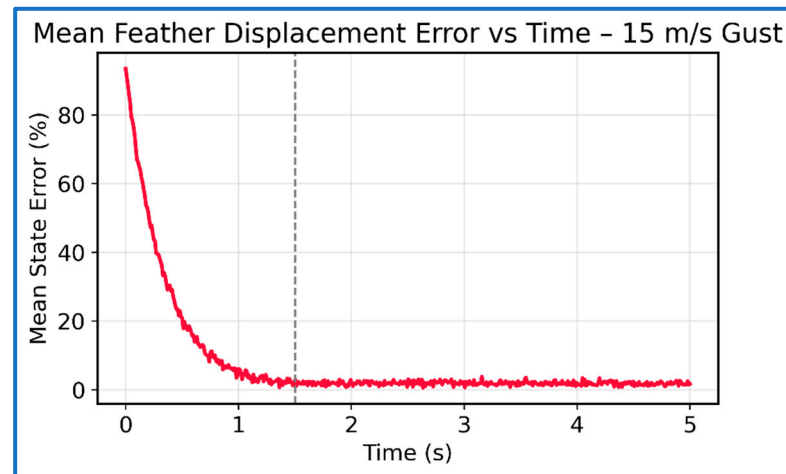


Figure 12. Mean state error at 15 m/s gust.

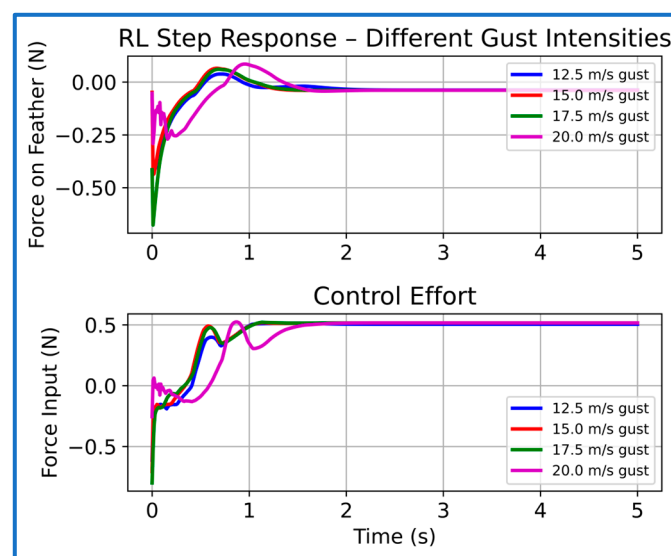


Figure 13. Step response and control effort of the DDPG agent at different gust intensities.

Table 3. Step response characteristics of the DDPG agent at different gust speeds: RT (Rise Time) and ST (Settling Time).

Gust (m/s)	RT (s)	ST (s)
12.5	0.01	1.52
15.0	0.00	1.25
17.5	0.01	1.24
20.0	0.00	1.59

To further assess robustness under time-varying disturbances, the trained controller was evaluated under a sinusoidal gust input with an amplitude of 10 m/s and 15 m/s. The results, shown in Figure 14, indicate that the controller maintains stable and bounded responses, with consistent force profiles. This demonstrates that the learned policy generalizes beyond step-type disturbances and remains effective under more realistic, continuously varying gust conditions.

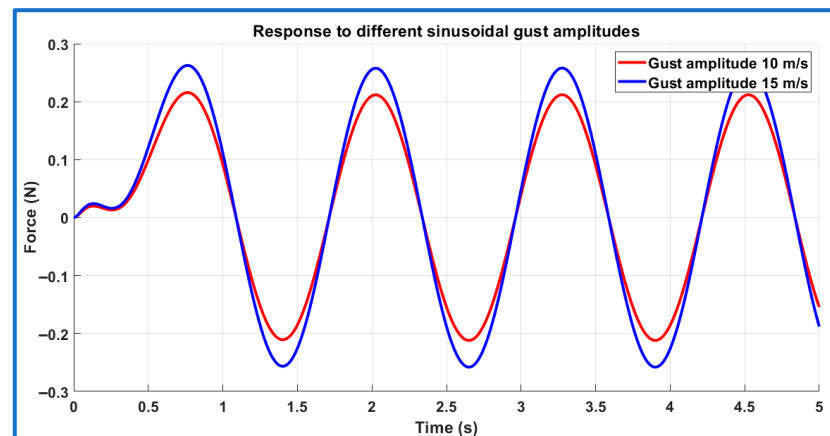


Figure 14. Response of the DDPG agent at different sinusoidal gust amplitudes.

The model was further simulated under two different gust intensities (15 m/s and 10 m/s) applied to two representative feathers (MCF 1 and MCF 3). Figure 15 presents the aerodynamic force acting on the flaps of these MCFs, generated by the voice-coil actuators in response to applied gust disturbances. Figure 16 illustrates the corresponding velocity profiles of these flaps. It was observed that both the force and velocity responses remain bounded and consistent across different gust intensities. Specifically, the force values are limited to approximately 0.53 N, while the feather velocities remain within actuator-feasible ranges (below 30 m/s). These values are within typical ranges for small-scale flapping-wing systems under moderate-to-high gust conditions. This further supports the physical validity of the proposed reduced-order model for control-oriented analysis.

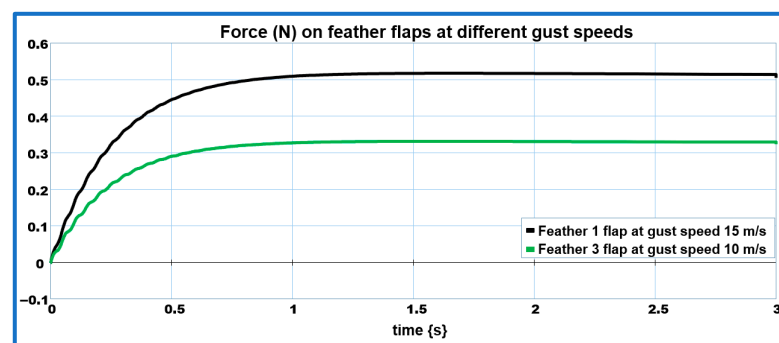


Figure 15. Force acting on feather flaps at different gust speeds.

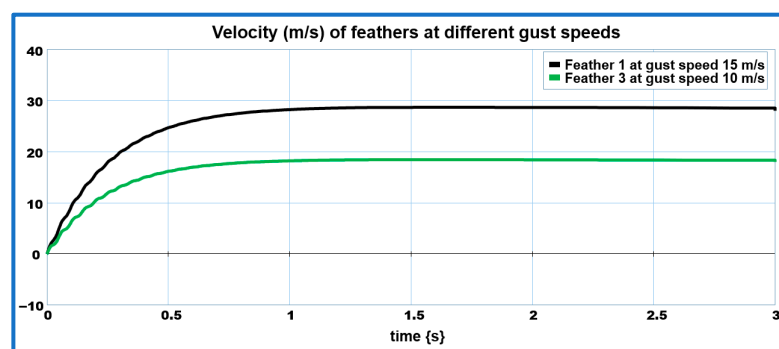


Figure 16. Velocity of feather flaps at different gust speeds.

Figure 17 illustrates the vertical displacement of the FWD wing at a 15 m/s gust, comparing the baseline configuration without an AFCU to the AFCU-equipped wing

controlled using a DDPG-based deep reinforcement learning strategy. The simulation shows that the activation of MCFs in the AFCU successfully suppresses gust by 50%. The developed DDPG-based agent additionally improves the stabilization speed, control efficiency and robustness. This validates the effectiveness of the current AFCU equipped with a DDPG-based DRL controller in handling gusts.

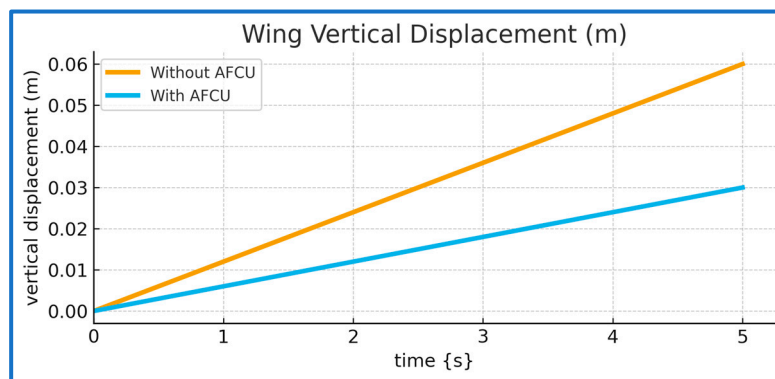


Figure 17. Response of wing with and without AFCU at 15 m/s gust.

The proposed DDPG-based controller’s efficacy is further supported by the comparison study in Table 4. The suggested approach maintains stable performance under gusts up to 20 m/s, which is the maximum disturbance level observed among the studied works, in contrast to earlier studies that showed gust tolerance limited to 7 m/s.

Table 4. Evaluation of the present DDPG controller with previous works.

Study	Technique	Disturbance	Results	Settling Time
Yu et al. [21]	PD and RL	Flips and aggressive maneuvers	Stable post-disturbance recovery and angular acceleration attenuation	≈1.5 s
Kim et al. [13]	Strain-based sensing + RL	3–7 m/s	Adaptive flight with increased stability	Converges to the target within approximately 5 s
Bhatia et al. [24]	LQR	Up to 3 m/s	Enhanced attitude stability with fewer oscillations	Converges within 1.8 s
Proposed	DDPG-Based DRL	Up to 20 m/s	Up to 50% gust suppression	Settling within 1.6 s

It should be noted that model-based analysis within the created simulation environment is the main emphasis of the validation carried out in this study. This physics-based representation of aero-mechanical dynamics was used to evaluate the DRL controller. The reduced-order model was developed from the full-order bond graph formulation of the AFCU system. Although early-stage control development for bio-inspired aerial systems frequently uses such model-based validation, experimental or high-fidelity CFD validation would offer more insight into real-world performance.

Furthermore, rather than implying direct experimental validation, the comparisons with previously published flapping-wing research are meant to offer qualitative context regarding the anticipated dynamic behavior of gust mitigation systems. The cited research is solely utilized to demonstrate consistency with trends seen in earlier experimental investigations of bio-inspired flight systems, while the current work assesses the suggested AFCU control framework inside a reduced model-based simulation environment. Hardware implementation and specialized wind tunnel testing would be necessary for a thorough

quantitative comparison with experimental platforms, which is still a crucial area for future research.

6. Conclusions

This study presented a peregrine falcon-inspired Active Flow Control Unit (AFCU) equipped with mechanical covert feathers (MCFs) and a Deep Deterministic Policy Gradient (DDPG)-based deep reinforcement learning (DRL) controller for active gust rejection in flapping-wing drones (FWDs). The reduced-order bond graph model effectively captured the nonlinear coupling between the main wing and feathered surfaces, forming the foundation for closed-loop DDPG training. The trained DDPG agent successfully stabilized the internally unstable open-loop AFCU, achieving near-zero steady-state error, bounded state trajectories, and smooth control efforts across step gust magnitudes ranging from 12.5 to 20 m/s. The resulting controller exhibited rapid convergence with settling times below 1.6 s, ensuring consistent aerodynamic stability and robustness under both moderate and severe gust conditions. Furthermore, the curriculum-based training results demonstrate stable and bounded responses under varying sinusoidal gust inputs, indicating improved robustness of the learned controller. In addition, the aerodynamic force and flap velocity responses remain within physically realistic and actuator-feasible limits, further supporting the practical relevance of the proposed approach. Simulation results further show up to 50% gust reduction compared to conventional methods. These results demonstrate that the proposed bio-inspired AFCU design augmented with DDPG-based DRL controller framework can autonomously learn optimal control strategies without predefined control laws, delivering adaptive gust suppression, energy-efficient actuation, and enhanced aerodynamic resilience compared to conventional gust alleviation methods that use traditional optimal or robust controllers.

In future studies, the presented control framework will be extended to include aerodynamic validation using CFD simulations to better understand how the MCFs interact with stochastic gust flows. In addition, hardware-in-the-loop experiments will be carried out to test the trained controller on an actual flapping-wing prototype and to study actuator response, communication delay, and power limits. Finally, we aim to explore transfer learning so that the trained model can be applied to other types of FWDs and to address model uncertainties and parameter variations that may appear during real flight tests. These steps will bring the proposed AFCU closer to practical application in real-world aerial environments.

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References

1. Liu, C.; Shen, T.; Shen, H.; Lu, B.; Sun, L.; Chen, G.; Chi, W. Mimicking Nature's Insects: A Review of Bio-Inspired Flapping-Wing Micro Robots (FWMRs). *J. Bionic Eng.* **2025**, *22*, 458–479. [\[CrossRef\]](#)
2. Chan, W.-L.; Liu, D.-J.; Chen, H.-Y.; Chin, C.-L. A Fan-Array Robotic-Arm Approach to Characterization of Pitch-Rate Dynamics of a Flapping-Wing MAV. *Actuators* **2025**, *14*, 592. [\[CrossRef\]](#)
3. Mohamed, A.; Marino, M.; Watkins, S.; Jaworski, J.; Jones, A. Gusts Encountered by Flying Vehicles in Proximity to Buildings. *Drones* **2023**, *7*, 22. [\[CrossRef\]](#)
4. Ren, Z.; Fu, W.; Yan, J. Gust Perturbation Alleviation Control of Small Unmanned Aerial Vehicle Based on Pressure Sensor. *Int. J. Aerosp. Eng.* **2018**, *2018*, 7259363. [\[CrossRef\]](#)
5. Ramos-Pedroza, N.; MacKunis, W.; Reyhanoglu, M. Synthetic Jet Actuator-Based Aircraft Tracking Using a Continuous Robust Nonlinear Control Strategy. *Int. J. Aerosp. Eng.* **2017**, *2017*, 4934281. [\[CrossRef\]](#)
6. De Rosa, D.; De Michele, C.; Catalano, P. Impact of Micro Vortex Generators on Aerodynamic Performance. *AIAA Pap.* **2025**, *2025*, 1069. [\[CrossRef\]](#)
7. Wang, S.; Li, Y.; He, W. Flight Attitude Estimation for MAV Based on M-Estimation. In *Proceedings of the International Conference on Consumer Electronics, Communications and Networks (CECNet)*; IEEE: New York, NY, USA, 2011; pp. 4968–4973. [\[CrossRef\]](#)
8. Tejaswi, K.C.; Kang, C.-K.; Lee, T. Dynamics and Control of a Flapping Wing UAV with Abdomen Undulation Inspired by Monarch Butterfly. In *Proceedings of the American Control Conference (ACC)*; IEEE: New York, NY, USA, 2021; pp. 66–71. [\[CrossRef\]](#)
9. Satoh, Y.; Ohno, K. Robust Attitude Stabilization of Rigid Bodies Based on Control Lyapunov Function: Experimental Verification on a Quadrotor Testbed. *Actuators* **2025**, *14*, 509. [\[CrossRef\]](#)
10. Wu, C.; Xiao, Y.; Zhao, J.; Mou, J.; Cui, F.; Liu, W. A Multi-Modal Tailless Flapping-Wing Robot Capable of Flying, Crawling, Self-Righting and Horizontal Take-Off. *IEEE Robot. Autom. Lett.* **2024**, *9*, 4734–4741. [\[CrossRef\]](#)
11. Wang, H.; Farid, Y.; Wang, L.; Garone, E.; Preumont, A. Hovering Flight of a Robotic Hummingbird: Dynamic Observer and Flight Tests. *Actuators* **2024**, *13*, 91. [\[CrossRef\]](#)
12. Coleman, D.; Benedict, M. Methods for Stabilizing the Longitudinal Dynamics of a Biomimetic Robotic Hummingbird in Hovering Flight. *Int. J. Micro Air Veh.* **2025**, *17*, 1–17. [\[CrossRef\]](#)
13. Kim, T.; Hong, I.; Im, S.; Rho, S.; Kim, M.; Roh, Y.; Kim, C.; Park, J.; Lim, D.; Lee, D.; et al. Wing-Strain-Based Flight Control of Flapping-Wing Drones through Reinforcement Learning. *Nat. Mach. Intell.* **2024**, *6*, 992–1005. [\[CrossRef\]](#)
14. Abbasi, S.H.; Mahmood, A. Bio-Inspired Gust Mitigation System for a Flapping Wing UAV: Modeling and Simulation. *J. Braz. Soc. Mech. Sci. Eng.* **2019**, *41*, 524. [\[CrossRef\]](#)
15. Hamzeh, F.; Fathollahi Dehkordi, S.; Naeimifard, A.; Abyaz, A. Continuously Variable Geometry Quadrotor: Robust Control via PSO-Optimized Sliding Mode Control. *Actuators* **2025**, *14*, 308. [\[CrossRef\]](#)
16. Ma, Q.; Wu, Y.; Shoukat, M.U.; Yan, Y.; Wang, J.; Yang, L.; Yan, F.; Yan, L. Deep Reinforcement Learning-Based Wind Disturbance Rejection Control Strategy for UAV. *Drones* **2024**, *8*, 632. [\[CrossRef\]](#)
17. Xiao, C.; Tang, L.; Wang, F.; You, S.; Xu, H.; Chen, M.; Lu, Z. Event-Triggered Control for Flapping-Wing Robot Aircraft System Based on High-Gain Observers. *Actuators* **2025**, *14*, 190. [\[CrossRef\]](#)
18. Dong, C.; Huo, Y.; Hui, N.; Han, X.; Tu, B.; Wang, Z.; Zhang, J. Decoupling-Free Attitude Control of UAV Considering High-Frequency Disturbances: A Modified Linear Active Disturbance Rejection Method. *Actuators* **2025**, *14*, 504. [\[CrossRef\]](#)
19. Yu, Y.; Lu, Q.; Zhang, B. Reinforcement Learning-Based Recovery Flight Control for Flapping-Wing Micro-Aerial Vehicles under Extreme Attitudes. *Int. J. Adv. Robot. Syst.* **2025**, *2025*, 22. [\[CrossRef\]](#)
20. Qian, C.; Fang, Y.; Li, Y. Neural Network-Based Hybrid Three Dimensional Position Control for a Flapping Wing Aerial Vehicle. *IEEE Trans. Cybern.* **2023**, *53*, 6095–6108. [\[CrossRef\]](#)
21. Xue, Y.; Cai, X.; Xu, R.; Liu, H. Wing Kinematics-Based Flight Control Strategy in Insect-Inspired Flight Systems: Deep Reinforcement Learning Gives Solutions and Inspires Controller Design in Flapping MAVs. *Biomimetics* **2023**, *8*, 295. [\[CrossRef\]](#)
22. Zheng, H.; Chen, W.; Xie, F. Control Simulation of Flapping Wing Micro Aerial Vehicle Based on Multi-Level Optimization Model Predictive Control. *IEEE Access* **2024**, *12*, 40700–40709. [\[CrossRef\]](#)
23. Dhingra, D.; Kaheman, K.; Fuller, S.B. Modeling and LQR Control of Insect-Sized Flapping Wing Robot. *NPJ Robot.* **2025**, *3*, 6. [\[CrossRef\]](#)

24. Bhatia, M.; Patil, M.; Woolsey, C. Stabilization of Flapping-Wing Micro-Air Vehicles in Gust Environments. *J. Guid. Control Dyn.* **2014**, *37*, 367–376. [[CrossRef](#)]
25. Zheng, Z.; Xu, X.; Wang, J.; Chen, Y.; Huang, J.; Wu, R.; Yu, H.; Lu, G. Tailless Flapping-Wing Robot with Bio-Inspired Elastic Passive Legs for Multi-Modal Locomotion. *IEEE Robot. Autom. Lett.* **2025**, *10*, 7971–7978. [[CrossRef](#)]
26. Rafee Nekoo, S.; Rashad, R.; De Wagter, C.; Fuller, S.B.; de Croon, G.; Stramigioli, S.; Ollero, A. A Review on Flapping-Wing Robots: Recent Progress and Challenges. *Int. J. Robot. Res.* **2025**, *44*, 2305–2339. [[CrossRef](#)]
27. Bechert, D.W.; Bruse, M.; Hage, W.; Meyer, R. Biological Surfaces and Their Technological Application—Laboratory and Flight Experiments on Drag Reduction and Separation Control. In Proceedings of the 28th AIAA Fluid Dynamics Conference, Snowmass Village, CO, USA, 29 June–2 July 1997. [[CrossRef](#)]
28. Abbasi, S.H.; Mahmood, A.; Khaliq, A. Bioinspired feathered flapping wing UAV design for operation in gusty environment. *J. Robot.* **2021**, *2021*, 8923599. [[CrossRef](#)]
29. Abbasi, S.H.; Mahmood, A.; Khaliq, A. Multi-Body Bond Graph Modeling and Simulation of a Bio-Inspired Gust Mitigating Flapping Wing UAV. *Songklanakarin J. Sci. Technol.* **2022**, *44*, 1238–1247.
30. Abbasi, S.H.; Mahmood, A.; Khaliq, A.; Imran, M. Reduced Order Modeling and Simulation of a Bio-Inspired Gust Mitigating Flapping Wing UAV. *Int. J. Intell. Robot. Appl.* **2022**, *6*, 587–601. [[CrossRef](#)]
31. Send, W.; Fische, M.; Jehens, K.; Mugrauer, R.; Nagarathinam, A.; Scharstein, F. Artificial Hinged-Wing Bird with Active Torsion and Partially Linear Kinematics. In Proceedings of the 28th International Congress of the Aeronautical Sciences (ICAS), Brisbane, Australia, 23–28 September 2012.
32. Gawthrop, P.J.; Bevan, G.P. Bond-Graph Modeling. *IEEE Control Syst. Mag.* **2007**, *27*, 24–45. [[CrossRef](#)]
33. Jahanbin, Z.; Ghafari, A.S.; Ebrahimi, A.; Meghdari, A. Multibody Simulation of a Flapping-Wing Robot Using an Efficient Dynamical Model. *J. Braz. Soc. Mech. Sci. Eng.* **2016**, *38*, 133–149. [[CrossRef](#)]
34. Samar, R.; Postlethwaite, I.; Gu, D.W. Model reduction with balanced realizations. *Int. J. Control* **1995**, *62*, 33–64. [[CrossRef](#)]
35. Lillicrap, T.P.; Hunt, J.J.; Pritzel, A.; Heess, N.; Erez, T.; Tassa, Y.; Silver, D.; Wierstra, D. Continuous Control with Deep Reinforcement Learning. *arXiv* **2019**, arXiv:1509.02971.
36. Silver, D.; Lever, G.; Heess, N.; Degris, T.; Wierstra, D.; Riedmiller, M. Deterministic Policy Gradient Algorithms. In Proceedings of the 31st International Conference on Machine Learning (ICML 2014), Beijing, China, 21–26 June 2014; pp. 387–395. Available online: <https://proceedings.mlr.press/v32/silver14.html> (accessed on 13 November 2025).

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